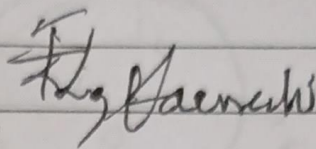


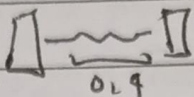
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Jawaban

$$x_{\text{natural}} = 40 \text{ cm} \\ = 0,4 \text{ m}$$



$$F = k \Delta x$$

$$\Rightarrow 60 = k \cdot (0,1)$$

$$\Rightarrow k = 600 \text{ N.m}^{-1}$$

Maka, di

$$W = \int_0^{\Delta x} F \cdot dx$$

$$= \frac{1}{2} k x^2 \Big|_0^{0,1}$$

$$= \frac{1}{2} \cdot 600 \cdot 0,01$$

$$= \underline{\underline{3 \text{ Joule}}}$$

$$\Delta x = x_{\text{natural}} - x_{\text{akhir}}$$

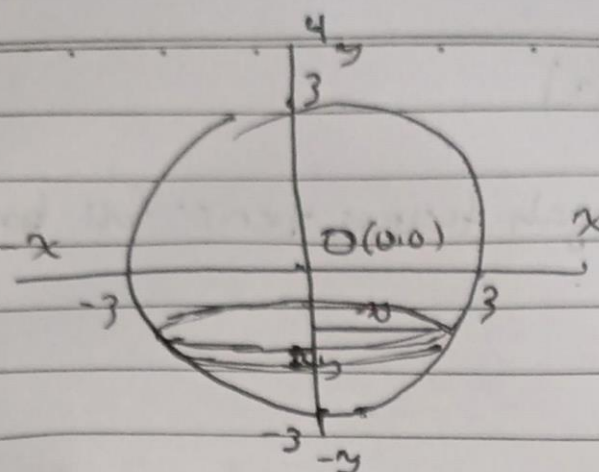
$$= 0,4 - 0,25$$

$$= 0,15$$

$$W = \frac{1}{2} k \cdot \Delta x^2$$

$$= \frac{1}{2} \cdot 600 \cdot (0,15)^2$$

$$= \underline{\underline{6,75 \text{ J}}}$$



$$\rho = 1000 \text{ kg/m}^3$$

$$\text{pakai } g = 10 \text{ m/s}^2$$

$$\pi r^2 \cdot h$$

$$\pi (g - y^2) \cdot (4 - y)$$

$$x^2 + y^2 = 9 \Rightarrow x^2 = 9 - y^2$$

$$\text{Angkat ke } y = 4 \Rightarrow h = 4 - y$$

$$\Delta W = F \cdot \Delta s$$

$$= \Delta m \cdot a \cdot \Delta s = a \cdot \Delta m \cdot \Delta s$$

$$= 10 \cdot \Delta V \cdot \rho \cdot \Delta s$$

$$= 10000 \cdot \pi (g - y^2) (4 - y)$$

Dengan demikian

$$W = \int_{-3}^3 F \cdot \Delta s = \int_{-3}^3 \Delta W$$

$$= \int_{-3}^3 10000 \pi (g - y^2) (4 - y) dy$$

$$= 10000 \pi \int_{-3}^3 (36 - \overset{\text{garis II}}{9y} - \overset{\text{garis I}}{4y^2} + y^3) dy$$

$$= 10000 \pi \int_{-3}^3 36 - 4y^2 dy$$

$$= 1000 \pi \left( 36y - \frac{4}{3} y^3 \right) \Big|_{-3}^3$$

$$= 10000 \pi (144)$$

$$= 1.440.000 \pi \text{ Joule}$$

$$\approx \boxed{4.524 \times 10^6 \text{ joule}}$$

Cara II (pusat massa)

$$W = m \cdot g \cdot \Delta h$$

pusat massanya di tengah karena terdiri dari bola yang  
oleh air.

$$\Delta h = R + d$$

$$= 4 \text{ m}$$

$$m = \rho \cdot V$$

$$= 1000 \frac{4\pi R^3}{3}$$

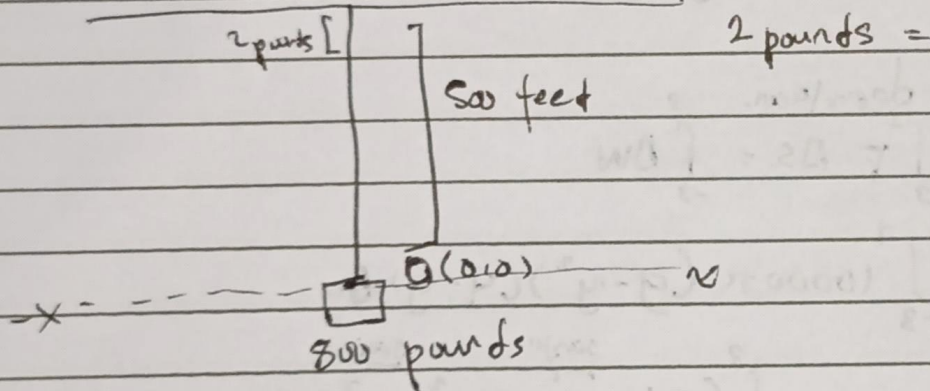
$$= 36000\pi \text{ kg}$$

Maka,

$$W = 36000\pi \times 10 \times 4$$

$$= 1.490.000\pi \text{ joule}$$

3.



$$W_{\text{total}} = W_{\text{tali}} + W_{\text{batu}}$$

$W_{\text{tali}}$  uniform sehingga bisa memakai pusat massa,  
yaitu

$$W_{\text{tali}} = F_{\text{tali}} \cdot h$$

$$= 2 \text{ pounds} \cdot 500 \text{ feet} \cdot 250 \text{ feet}$$

$$= 250.000 \text{ ft-lbs}$$

(Bukti pakai integral)

$$\begin{aligned}
 W_{\text{tarik}} &= \int_0^{500} F_{\text{tarik}} \cdot ds \\
 &= \int_0^{500} y \cdot 2 \, dy \\
 &= y^2 \Big|_0^{500}
 \end{aligned}$$

$$= 250.000 \text{ ft-lbs}$$

Dari energi potensial, d. dapat

$$W_{\text{berat}} = m \cdot g \cdot h$$

$$= F \cdot d$$

$$= 800 \cdot 500$$

$$= 400.000 \text{ ft-lbs}$$

$$\therefore \boxed{W_{\text{total}} = 650.000 \text{ ft-lbs}}$$

$$= \boxed{881.270 \text{ joule}}$$

$$f(x) = \ln(x-1) + 1$$

$\Rightarrow$   $\ln$  yang digeser ke kanan satu unit dan 1 unit ke atas.

tidak berpengaruh terhadap Df.

Densitas adalah  $\{x \mid x > 0, x \in \mathbb{R}\}$

Sehingga jika digeser ke kanan,  $x'$  baru adalah

$$x' - 1 = 0 \Rightarrow x' = 1$$

$$\text{Maka, } \boxed{D_{f(x)} = \{x \mid x > 1, x \in \mathbb{R}\}}$$

$$\text{Potong sumbu-} x \Rightarrow y = 0$$

$$\Rightarrow 0 = \ln(x-1) + 1$$

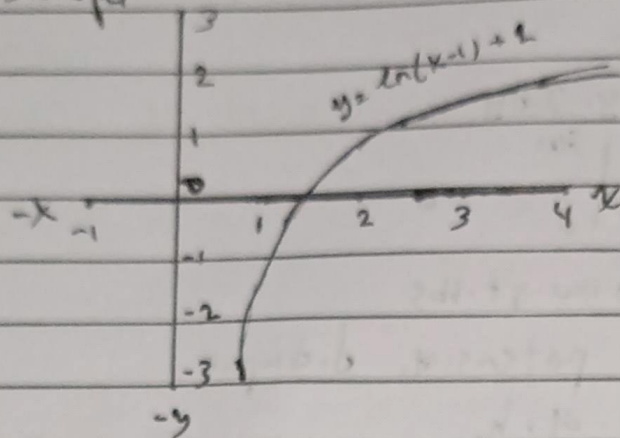
$$\Rightarrow -1 = \ln(x-1)$$

Pakai invers / definisi exp dengan  $e \approx 2.71828$

$$\Rightarrow \exp(-1) = x-1$$

$$\Rightarrow \boxed{x = 1 + \exp(-1)} \text{ atau } \boxed{x = 1 + \frac{1}{e}}$$

Grafiknya berupa



5. (a)  $\int_2^4 \frac{1}{x \ln x} dx$

Substitusi  $u = \ln x \Rightarrow u_0 = \ln(x_0) = \ln(2)$   
 $\Rightarrow du = \frac{1}{x} dx \quad u_1 = \ln(x_1) = \ln(4)$

$\Rightarrow \int_{\ln(2)}^{\ln(4)} \frac{1}{u} du$

$= \ln(u) \Big|_{\ln(2)}^{\ln(4)}$

$= \ln(\ln 4) - \ln(\ln 2)$

$= \ln(2 \ln 2) - \ln(\ln 2)$

$= \boxed{\ln 2}$

(b)  $\ln(xy) = e^{x+y}$ , turunkan

$\Rightarrow \frac{1}{xy} (y + y') = (1 + y') e^{x+y}$

$\Rightarrow \frac{1}{x} + \frac{y'}{y} = e^{x+y} + y' e^{x+y}$

$\Rightarrow \frac{1}{x} - e^{x+y} = y' e^{x+y} - \frac{y'}{y}$



8. Integral biasa dengan  $t(0) = 1, f(0) = 3$

$$f'(x) = \int f''(x) dx$$

$$= \int 4e^{2x} + \frac{1}{(x+1)^2} dx$$

$$= 2e^{2x} - \frac{1}{x+1} + C_0$$

Masukkan  $x=0$ ,

$$1 = 2 - 1 + C_0$$

$$\Rightarrow C_0 = 0$$

$$\Rightarrow f'(x) = 2e^{2x} - \frac{1}{x+1}$$

$$f(x) = \int f'(x) dx$$

$$= e^{2x} - \ln|x+1| + C_1$$

Masukkan  $x=0$

$$3 = 1 - 0 + C_2$$

$$\Rightarrow C_2 = 2$$

Dengan demikian,

$$\boxed{f(x) = e^{2x} - \ln|x+1| + 2}$$